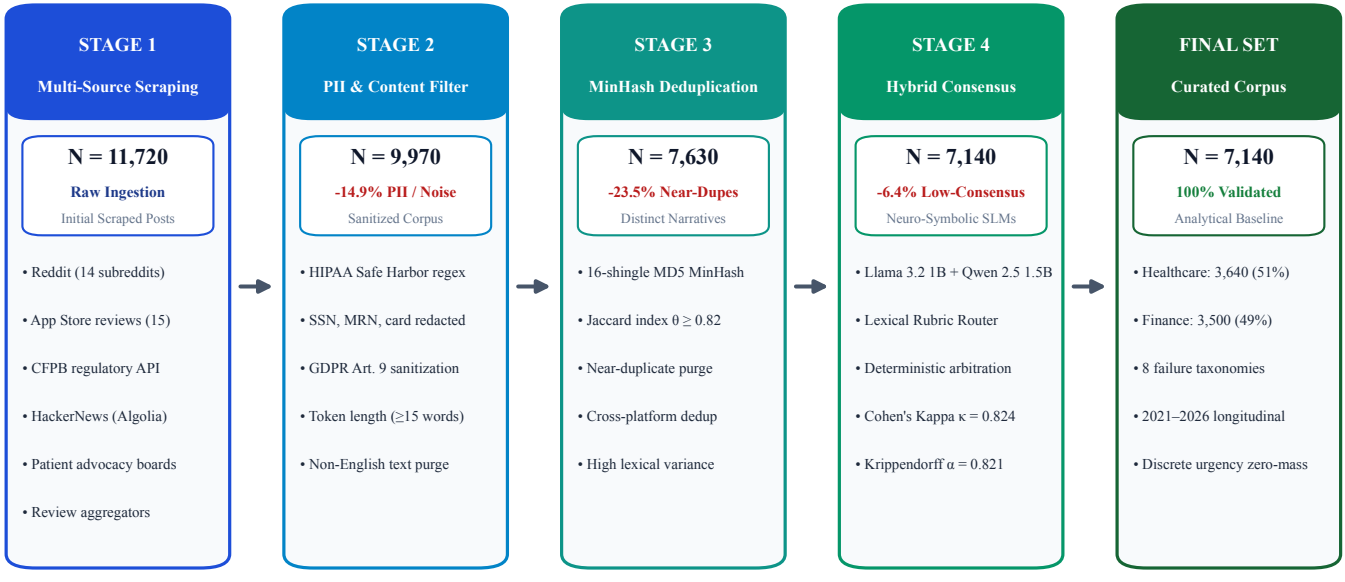

STAGE 1
Multi-Source Scraping
N = 11,720
Raw Ingestion
Initial Scraped Posts
• Reddit (14 subreddits)
• App Store reviews (15)
• CFPB regulatory API
• HackerNews (Algolia)
• Patient advocacy boards
• Review aggregators
STAGE 2
PII & Content Filter
N = 9,970
-14.9% PII / Noise
Sanitized Corpus
• HIPAA Safe Harbor regex
• SSN, MRN, card redacted
• GDPR Art. 9 sanitization
• Token length (≥15 words)
• Non-English text purge
STAGE 3
MinHash Deduplication
N = 7,630
-23.5% Near-Dupes
Distinct Narratives
• 16-shingle MD5 MinHash
• Jaccard index θ ≥ 0.82
• Near-duplicate purge
• Cross-platform dedup
• High lexical variance
STAGE 4
Hybrid Consensus
N = 7,140
-6.4% Low-Consensus
Neuro-Symbolic SLMs
• Llama 3.2 1B + Qwen 2.5 1.5B
• Lexical Rubric Router
• Deterministic arbitration
• Cohen's Kappa κ = 0.824
• Krippendorff α = 0.821
FINAL SET
Curated Corpus
N = 7,140
100% Validated
Analytical Baseline
• Healthcare: 3,640 (51%)
• Finance: 3,500 (49%)
• 8 failure taxonomies
• 2021–2026 longitudinal
• Discrete urgency zero-mass